

Static and Dynamic Binding

Binding is the process of connecting a **function call** to the **actual block of code** that executes it. (When we call a function like **(car.drive())**, it will connect it with the memory address of where the **drive()** function is written)

The difference between **Static** and **Dynamic binding** is entirely about **when** that connection happens.

1. Static Binding (Early Binding)

When it happens: before the program runs (during compilation).

What it is: The computer knows exactly which code to execute just by looking at the rules of the code. Everything is locked in early.

Advantage: It is very fast.

Analogy: Buying a ticket for a specific assigned seat at a concert day in advance.

2. Dynamic Binding (Late Binding)

When it happens: while the program is actively running (at runtime).

What it is: The computer figures out which code to execute at the exact moment it is called, based on the specific object being used right then.

Advantage: It is highly flexible.

Analogy: Walking into a cafe and choosing where to sit based on which tables are open at that exact moment.